

# Homework 3: Probability and Testing Independence

Algorystics Education | Data downloaded 3 October 2026

## 1. Two-dice simulation

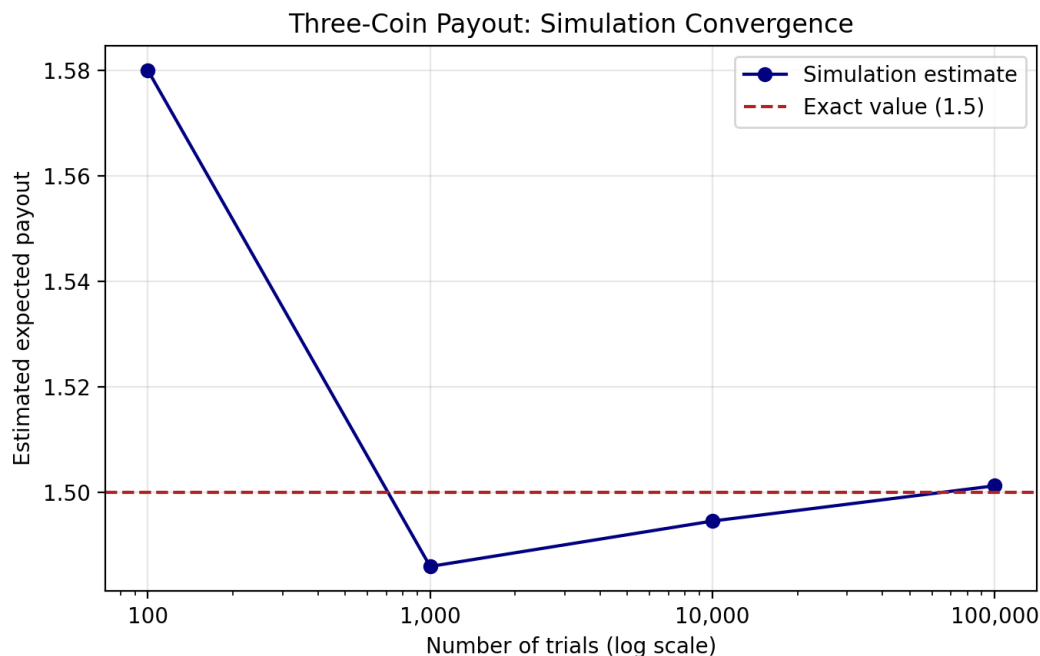
I used NumPy's random generator with fixed seed **42** and ran 100,000 trials. The estimate of **P(4-sided | rolled a 1)** was **0.6050**, compared with the exact answer **0.6000**, for a difference of **+0.0050**.

## 2. Three-coin payout

With 100,000 trials and seed 42, the simulated expected payout was **1.5013**. The exact expected payout is **1.5000**.

## 3. Simulation convergence

| Trials  | Estimated payout | Exact payout |
|---------|------------------|--------------|
| 100     | 1.5800           | 1.5000       |
| 1,000   | 1.4860           | 1.5000       |
| 10,000  | 1.4946           | 1.5000       |
| 100,000 | 1.5013           | 1.5000       |



The estimate becomes much more stable as trials increase. This is the practical reason simulation and backtest results need enough observations before they are trusted.

#### 4-6. Are daily SPY returns independent?

I downloaded 10 years of unadjusted daily SPY closes (3 October 2016 through 2 October 2026), yielding **2,514 trading days** and 2,513 daily returns. The mean daily return was **0.0571%**.

| Quantity                                   | Estimate | Count supporting estimate       |
|--------------------------------------------|----------|---------------------------------|
| P(tomorrow is down)                        | 0.4481   | 1,126 down days / 2,513 returns |
| P(tomorrow down   today down)              | 0.4414   | 1,126 qualifying down days      |
| P(tomorrow down   today down more than 2%) | 0.4023   | 87 qualifying big-drop days     |

#### 7. Expected present value

**expected\_present\_value([10, 10, 10], 0.10, 0.5) = 15.10**. Year 1 is certain, Year 2 is weighted by 0.5, and Year 3 by 0.25 before each cash flow is discounted.

#### 8. Interpretation

These data do not provide strong evidence of a useful one-day dependence in the direction tested: the unconditional probability of a down day is 44.81%, while the probability after a down day is 44.14%. After a drop greater than 2%, the estimate is lower at 40.23%, but it is based on only 87 days, so that difference is much less precise. Thus, independence is a reasonable rough approximation for this simple next-day test, but it is not proven by it; returns can still have changing volatility, tail risk, or patterns at other horizons. A strategy that assumes independent daily returns should therefore test its signal out of sample and stress its results rather than treating small conditional-probability differences as a reliable trading edge.